

Student Management System

1. What is CRUD in FastAPI?

CRUD stands for Create, Read, Update, and Delete operations. These operations are used to manage data in an API.

2. Why do we use the POST method for inserting data?

POST method is used to send new data to the server. It helps in creating new records in the database.

3. Difference between PUT and PATCH

PUT updates the complete data of a resource. PATCH updates only specific fields of a resource.

4. What is the path parameter?

A path parameter is a value passed in the API URL. It is used to identify a specific resource like student ID.

5. What is the request body in FastAPI?

Request body contains the data sent by the client to the API. FastAPI uses it to receive JSON data for POST and PUT methods.

6. Why do we use Pydantic models?

Pydantic models are used for data validation and type checking. They ensure correct data is received in the API.

7. What status code is returned for successful deletion?

The common success status code for deletion is 200 OK. Sometimes 204 No Content is also used.

8. How can we check whether a student exists before updating?

We check the student ID in the students list before updating. If the ID is not found, FastAPI returns an error message.

9. What happens if duplicate IDs are inserted?

Duplicate IDs can create data inconsistency and confusion. We should validate IDs before inserting new records.

10. How does FastAPI automatically generate Swagger UI?

FastAPI automatically creates Swagger UI from API routes and models. It provides interactive API documentation at [/docs](#).